

Supplement to Planning Shared Review for Language Model Workflows with Expiring Intervention Opportunities

Anonymous Companion Material

Keywords: Reproducibility, Review Semantics, Paired Evaluation.

Abstract: This companion records experimental matrices, operational assumptions, source-case provenance, paired robustness diagnostics and replay procedures for the shared-review study. It separates the original maintenance distributions from constructed competition, and separates fresh retail source cases from post hoc analysis of earlier preparations. Detailed outcome tables retain failures and blocked unresolved requests. The main paper contains the formulation, central comparisons, uncertainty and limitations needed to assess its claims. This supplement provides additional inspection and reproduction detail. It is a local research companion and is not represented as an accepted supplementary submission under the conference policy.

1 EVIDENCE ORGANIZATION

The evidence package retains immutable declarations, source snapshots, model requests and responses, event traces, estimators and accounting ledgers. Development chronology resides in stage reports rather than the main scientific argument. Original, competition and larger maintenance distributions remain separate. Original retail and the fresh operational follow-up are also separate cohorts.¹

The completed maintenance evaluation contains 2,752 episodes, with 304 development episodes and 128 separate repeated-request diagnostic generations. Its 48,320 GPU attempts include all development, evaluation and diagnostic calls. The original retail stage contains two preserved pilots, a 72-workflow calibration set and a 288-workflow evaluation. The first pilot staged no transactions; one declared interface revision corrected tool-before-argument serialization and added public progress information. It does not identify the separate causal effects of those two changes. The frozen follow-up leaves this interface unchanged.

1.1 Maintenance Matrices

Original scenarios use seeds 10000–10063 and competition uses 11000–11063. Each has six policies and both review durations under frozen agreement risk. The two alternative-risk rules use the first 32 seeds in each workload with greedy and search,

while reusing matching frozen rows as the reference. Larger-workload scenarios use 12000–12031, three and six agents, four policies and both durations. The selected objective extension uses 13000–13015 with the declared additional-closure penalties 0, 4 and 8. It is a development-selected objective study, not a universal objective search.

The original distribution and competition matrices contribute 1,536 episodes, alternative-risk rollouts 512, larger workload 512, and the objective extension 192. These are repeated conditions, not 2,752 independent scenarios. The frozen scenario prefixes are not extended based on observed outcomes. All completed traces replay.

1.2 Observation and Feedback Boundary

Maintenance model inputs contain released public job information and that agent’s own history. Schedulers receive public request records rather than private job objects. Before review completion, neither has definitive fault labels. Reviews finish before same-tick deadline closure, so equality can prevent a terminal penalty. Downtime accumulated before correction remains. Search has no access to the generator’s future waves.

Retail preparation has its own agent history and no access to annotated targets. Authentication, account ownership, retrieval and confirmation checks apply before a mutation can be staged. The scheduler then receives only arrived transaction requests and public risk, cost, cutoff and service-time fields. Hidden

¹Drafting assistance: OpenAI Codex (OpenAI, 2026). See the AI assistance disclosure.

proposal correctness is consulted by completed perfect review and offline scoring. Other-policy choices logged on the same public state are diagnostics and do not alter the executed policy.

2 RETAIL SOURCE AND SEMANTICS

The source is the MIT-licensed retail component of τ -bench (Yao et al., 2024), pinned at revision

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The selected substrate retains all orders for each isolated account and the full product catalog. Tool functions are unmodified. The original package’s hosted user simulator is not imported. The adapter reproduces its recursive final-database hash semantics and verifies target mutations on isolated state.²

The supported mutation tools are cancellation, pending-order item modification, delivered-order item return and delivered-order item exchange. Account lookup, account retrieval, order retrieval and product retrieval remain genuine tool steps. Item-modification and exchange calls require the full desired list, compatible available variants and valid payment. A wrong but available variant can pass these checks and fail the recorded user intent. Review therefore addresses a different error class from an invalid tool call.

The customer script repeats only the task’s operational information. It confirms the proposal that the agent displays, without independently checking product identifiers against hidden targets. This is not an oracle user. The source audit checks intent-to-annotation consistency without model outputs, including order identity, cancellation reason, refund/payment destination, old/new options, unique available target variants, return names and multiplicities. Broad source templates remain shared across partitions.

2.1 Operational Assumptions

The following distinctions guide interpretation.

- A staged mutation is a proposed database operation. Posting it is a committed application transition. Physical delivery, receipt and payment settlement are not simulated.
- A review cutoff is our constructed latest completion time for preventing the posting consequence.

²Drafting assistance: OpenAI Codex (OpenAI, 2026). See the AI assistance disclosure.

Table 1: Frozen application calibration. All staged examples have the uncertainty flag. Unflagged bins have no examples and use pooled fallback.

Family	Examples	Errors	Risk
Cancellation	23	0	0.0400
Modification	22	6	0.2917
Return/exchange	19	9	0.4762
Pooled	64	15	0.2424

It is not a measured retailer deadline. The backend does not advance a pending order to shipped merely because a tick passes.

- Corrective review assumes permission to restore the complete recorded intent without renewed customer confirmation. Restricted review removes that power and retains perfect detection only.
- Blocking a wrong transaction prevents its processing consequence but leaves four points of unresolved service loss. It is neither a completed task nor a successful correction.
- Complexity time depends on the proposed public item list. If an agent omits an item, the proxy may make that proposal appear simpler. This is retained rather than repaired using hidden targets.
- Exact database equality may penalize a reason or payment-field difference more strictly than a domain expert’s operational equivalence judgment. Such judgments were not collected.

Shopify’s documented editing, cancellation and return-processing interfaces support the distinction between request, approval and later processing (Shopify, 2026b; Shopify, 2026a; Shopify, 2026c). They do not establish the simulation’s numerical weights, perfect detection or service times. The source-based methodological review is not independent human expert validation.

2.2 Risk Estimator

Calibration includes all 64 successfully staged proposals from 72 development workflows, with labels taken before correction. No reviewed-only label filter is applied. Let a bin have e initial errors among n proposals. Its estimate is $(e + 1)/(n + 2)$ if $n \geq 10$, otherwise the pooled calibration estimate. The public feature is family plus an uncertainty flag for non-high reported confidence or a previous automatic rejection.

The frozen estimator hash is

5734f6babfbd33655a76595f001251bd591
4d94dde0c0e3401af3fc4a762a581

No follow-up labels refit it. Prediction diagnostics condition on staging and are identical across scheduling conditions because preparations are shared. Unstaged failures remain in task and loss reporting.

3 FRESH FOLLOW-UP PROVENANCE

The source-pool audit examines all 635 pinned train, dev and test records. It finds 54 customer accounts appearing in multiple splits. Exclusion operates on account identity across splits, not just record IDs. All 120 Stage 5 accounts are excluded, including development and pilot records. Remaining eligible accounts number 10 for modification, 26 for cancellation and 69 for returns/exchanges before cross-family reservation. The 16-bundle target is therefore infeasible. The frozen fallback selects eight bundles from 24 globally unique training accounts. The source split does not establish absence from model training.³

Source order is train, dev, test, then ascending index. Modification is reserved first, followed by cancellation and returns/exchanges, with account deduplication throughout. No model outcome influences selection. Three generation replicates use identical case composition and public slot assignments. Scenario seeds 62000–62007 permute arrivals, windows, weights and agent-family order. Generation seeds hash the partition, bundle, case, replicate, step and retry, independently of policy or operational variant.

The full frozen follow-up is 72 workflow attempts and 864 policy episodes. Its maximum preparation cost is 1,008 ordinary calls or 2,016 attempts with one retry per call. The stage-wide ceiling is 6,000 attempts. Model settings, source hashes, estimator, conditions, comparisons and failure handling were committed before the first fresh call. No difficult case is replaced.

4 PAIRED ANALYSIS DETAILS

For each source bundle, the three generation replicates are averaged before comparing policies. Bootstrap draws resample entire bundles, with all conditions kept together. The fresh sample therefore has eight independent bundle units. The post hoc sensitivity uses 32 earlier bundles and is not pooled with it. Fixed seed 20260916 generates 2,000 common-index

³Drafting assistance: OpenAI Codex (OpenAI, 2026). See the AI assistance disclosure.

resamples for the follow-up analyses.⁴

The original Stage 5 primary interval remains $[-0.3333, 1.0000]$ with its original analysis seed. Re-computing the same reference rows with the newly declared common bootstrap seed gives a slightly different upper endpoint, 0.9167. This finite Monte Carlo difference does not reflect changed outcomes. Both records are preserved.

A reporting-direction defect in secondary completion rows was detected and repaired. The initial code labeled a negative completion difference as a win because it inherited the loss convention. The corrected labels treat more completed tasks as better. Numeric differences, intervals, simulation outcomes and the primary loss comparison are unchanged; the original tables and patch hashes are retained. Frozen preparation code and source snapshots remain intact.

5 GENERATION AND PLANNING COSTS

Experiments use Qwen/Qwen2.5-7B-Instruct revision

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on one RTX 6000 Ada, BF16, with CPU model of-float and swap disabled. The reused runtime has Python 3.11.13, PyTorch 2.8.0 with CUDA 12.8, vLLM 0.10.2 and Transformers 4.55.2. The driver is 580.126.20. Driver-reported compatibility is not used as evidence of the installed runtime. CUDA BF16 computation, server arguments, model identity, process ancestry and GPU allocation are recorded together.⁵

The original server on port 8000 is preserved. A separate retail server uses a 16,384-token context and 0.4 GPU-memory reservation. Requests are serial. Tokenization runs before each generation and enforces the declared input bound. Each attempt has a 60-second timeout, and each ordinary call allows one retry. Every request and failed attempt is retained in a session ledger and reconciled with raw evidence and server counters.

Earlier maintenance exact search reached six eligible requests and 1,957 ordered subsets, with 22 observed decisions averaging 4.531 ms on its recorded host. Original retail search had at most two eligible requests at actual dispatch, examining five subsets and averaging 0.05135 ms over 218 such decisions. These

⁴Drafting assistance: OpenAI Codex (OpenAI, 2026). See the AI assistance disclosure.

⁵Drafting assistance: OpenAI Codex (OpenAI, 2026). See the AI assistance disclosure.

Table 2: Fresh bundle composition. Each row is reused across three generation replicates. Values are case ID, arrival, cutoff and processing weight in agent order.

Bundle	Agent 0	Agent 1	Agent 2
stage6_00	train:313 / 0 / 2 / 8	train:421 / 0 / 4 / 12	train:153 / 1 / 6 / 4
stage6_01	train:158 / 0 / 2 / 8	train:320 / 1 / 5 / 12	train:436 / 0 / 5 / 4
stage6_02	train:437 / 1 / 5 / 8	train:162 / 0 / 2 / 12	train:323 / 0 / 5 / 4
stage6_03	train:167 / 0 / 2 / 4	train:324 / 1 / 6 / 12	train:439 / 0 / 4 / 8
stage6_04	train:169 / 0 / 4 / 4	train:338 / 0 / 2 / 8	train:450 / 1 / 6 / 12
stage6_05	train:174 / 0 / 5 / 12	train:455 / 1 / 5 / 8	train:341 / 0 / 2 / 4
stage6_06	train:466 / 0 / 5 / 4	train:177 / 1 / 5 / 8	train:356 / 0 / 2 / 12
stage6_07	train:357 / 0 / 2 / 4	train:482 / 1 / 5 / 12	train:178 / 0 / 5 / 8

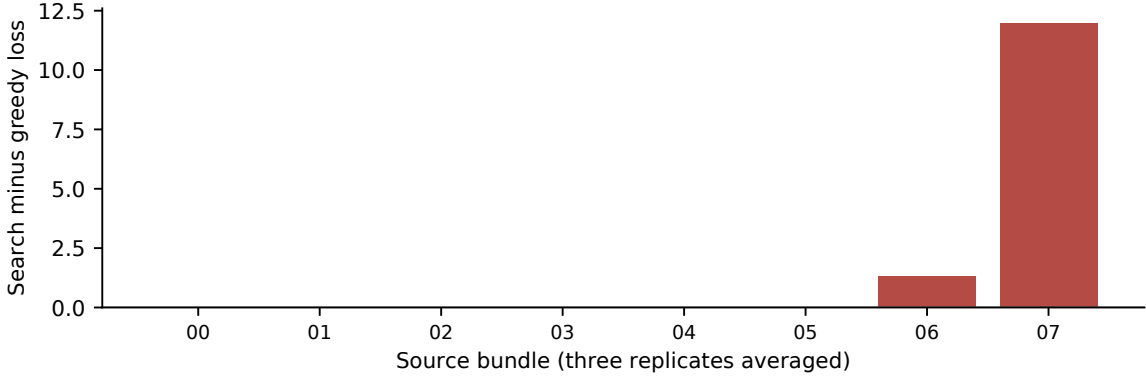


Figure 1: All eight fresh primary bundle differences, after averaging three generation replicates. Zero values remain explicit.

are bounded host-planning measurements, not a scalability guarantee or GPU latency measure.

The fresh follow-up completes 675 ordinary generation calls and 675 attempts, with zero retries, failures or unknown-token records. Requests consume 3,495,749 prompt tokens and 58,762 completion tokens. Summed HTTP generation time is 1,149.669 seconds; the preparation worker spans 1,200.660 seconds. The longest attempt lasts 4.993 seconds. Maximum observed input/output lengths are 11,369/256 tokens, within their declared limits. Server counter deltas agree exactly with the ledger and raw attempts.

There are no repeated byte-identical request groups within this fresh run. Different seeded replicates yield different proposals on eight of 24 source cases and mixed initial correctness on four. These comparisons involve different generation seeds and often later prompt histories; they do not measure identical-request nondeterminism.

Across fresh operational conditions, actual search dispatch reaches two eligible requests. Its 114 such decisions examine five ordered subsets and average 0.04832 ms, with maximum 0.11583 ms on the recorded host. Three pending requests can appear in no-review diagnostic states; the exact planner is not truncated. These repeated state measurements are descriptive.

The stage starts at 21:18:02 UTC on 10 September 2026, with inference cutoff 01:48:02 and hard deadline 03:18:02 the next day. The last generation completes at 21:57:00.893 UTC, well before the reporting reserve. The temporary server and both experimental worker processes are verified absent at 21:59:19 UTC. The original server remains healthy with unchanged counters. No new paid resources are provisioned. Provider allocation duration, price and charge are unavailable, so request time is not converted into billed cost. The final stage and cumulative clocks are reported in the repository report. Across both Stage 1 GPU demonstrations and Stages 2–6, the recorded total is 55,384 generation attempts, including two retries. No timestamp is reset to fit the original 36-hour target.

The 13,459 historical artifact, configuration and upstream files captured at authorization remain byte-identical. All 72 new GPU workflows, 864 fresh policy traces and 3,456 post hoc traces replay. The latter include 1,152 exact matches to historical retail reference traces. No fresh public evidence requires redaction; historical redaction provenance remains disclosed.

Table 3: Fresh cases. Absolute outcomes across three generation replicates per bundle. Totals are repeated-condition counts, not independent samples. S and W denote service and wrong-posting loss. B is blocked unresolved requests and U is unstaged failures.

Condition / base	Policy	Loss	Mean	S	W	Correct	Wrong	B	U	Missed
Correction / 1	No review	236	9.833	132	104	39	12	0	21	12
Correction / 1	FCFS	84	3.500	84	0	51	0	0	21	0
Correction / 1	EDF	84	3.500	84	0	51	0	0	21	0
Correction / 1	Uncertainty	84	3.500	84	0	51	0	0	21	0
Correction / 1	Greedy	84	3.500	84	0	51	0	0	21	0
Correction / 1	Search	84	3.500	84	0	51	0	0	21	0
Correction / 2	No review	236	9.833	132	104	39	12	0	21	12
Correction / 2	FCFS	132	5.500	96	36	48	3	0	21	3
Correction / 2	EDF	132	5.500	96	36	48	3	0	21	3
Correction / 2	Uncertainty	84	3.500	84	0	51	0	0	21	0
Correction / 2	Greedy	84	3.500	84	0	51	0	0	21	0
Correction / 2	Search	140	5.833	100	40	47	4	0	21	4
Approve/block / 1	No review	236	9.833	132	104	39	12	0	21	12
Approve/block / 1	FCFS	132	5.500	132	0	39	0	12	21	0
Approve/block / 1	EDF	132	5.500	132	0	39	0	12	21	0
Approve/block / 1	Uncertainty	132	5.500	132	0	39	0	12	21	0
Approve/block / 1	Greedy	132	5.500	132	0	39	0	12	21	0
Approve/block / 1	Search	132	5.500	132	0	39	0	12	21	0
Approve/block / 2	No review	236	9.833	132	104	39	12	0	21	12
Approve/block / 2	FCFS	168	7.000	132	36	39	3	9	21	3
Approve/block / 2	EDF	168	7.000	132	36	39	3	9	21	3
Approve/block / 2	Uncertainty	132	5.500	132	0	39	0	12	21	0
Approve/block / 2	Greedy	132	5.500	132	0	39	0	12	21	0
Approve/block / 2	Search	172	7.167	132	40	39	4	8	21	4
Complexity time / 1	No review	236	9.833	132	104	39	12	0	21	12
Complexity time / 1	FCFS	84	3.500	84	0	51	0	0	21	0
Complexity time / 1	EDF	84	3.500	84	0	51	0	0	21	0
Complexity time / 1	Uncertainty	84	3.500	84	0	51	0	0	21	0
Complexity time / 1	Greedy	84	3.500	84	0	51	0	0	21	0
Complexity time / 1	Search	84	3.500	84	0	51	0	0	21	0
Complexity time / 2	No review	236	9.833	132	104	39	12	0	21	12
Complexity time / 2	FCFS	140	5.833	100	40	47	4	0	21	4
Complexity time / 2	EDF	140	5.833	100	40	47	4	0	21	4
Complexity time / 2	Uncertainty	92	3.833	88	4	50	1	0	21	1
Complexity time / 2	Greedy	92	3.833	88	4	50	1	0	21	1
Complexity time / 2	Search	148	6.167	104	44	46	5	0	21	5

6 EXAMPLE SELECTION

Examples follow the frozen rule of the first numeric bundle and replicate satisfying each category under the primary restricted-authority condition. Categories are search benefit, tie, unfavorable result and preparation failure. A missing category is reported rather than filled with a convenient alternative. Full traces retain tool proposals, annotated targets for offline interpretation, review starts/completions, blocks, committed changes and incurred consequences.⁶

⁶Drafting assistance: OpenAI Codex (OpenAI, 2026). See the AI assistance disclosure.

6.1 Fresh Unfavorable and Tied Cases

No fresh restricted-authority run gives search lower loss than greedy. The first unfavorable run is bundle 06, replicate 2. Modification case train:466 proposes a wrong available variant. It has arrival zero, cutoff five, processing weight four and predicted error 0.2917. Cancellation case train:356 is correct, with arrival zero, cutoff two, weight twelve and predicted error 0.04. Exchange train:177 arrives at one, is correct, and has cutoff five, weight eight and predicted error 0.4762.

Search first protects the expiring cancellation, then chooses the exchange at tick two. The modification can no longer finish review after tick four and posts at five. Its service cost four plus posting cost four yields loss eight. Greedy blocks the modification

Table 4: Fresh cases. Review resources. Time and waiting are summed abstract ticks. Utilization is mean occupied fraction of the processing horizon. More reviews do not necessarily imply more useful interventions.

Condition / base	Policy	Reviews	Corrections	Time	Waiting	Utilization	Multi-eligible
Correction / 1	No review	0	0	0	0	0.000	57
Correction / 1	FCFS	51	12	51	19	0.401	19
Correction / 1	EDF	51	12	51	19	0.401	19
Correction / 1	Uncertainty	46	12	46	9	0.361	19
Correction / 1	Greedy	46	12	46	9	0.361	19
Correction / 1	Search	51	12	51	19	0.401	19
Correction / 2	No review	0	0	0	0	0.000	38
Correction / 2	FCFS	44	9	88	32	0.686	18
Correction / 2	EDF	44	9	88	33	0.686	19
Correction / 2	Uncertainty	42	12	84	23	0.658	14
Correction / 2	Greedy	42	12	84	23	0.658	14
Correction / 2	Search	44	8	88	32	0.686	19
Approve/block / 1	No review	0	0	0	0	0.000	57
Approve/block / 1	FCFS	51	0	51	19	0.401	19
Approve/block / 1	EDF	51	0	51	19	0.401	19
Approve/block / 1	Uncertainty	46	0	46	9	0.361	19
Approve/block / 1	Greedy	46	0	46	9	0.361	19
Approve/block / 1	Search	51	0	51	19	0.401	19
Approve/block / 2	No review	0	0	0	0	0.000	38
Approve/block / 2	FCFS	44	0	88	32	0.686	18
Approve/block / 2	EDF	44	0	88	33	0.686	19
Approve/block / 2	Uncertainty	42	0	84	23	0.658	14
Approve/block / 2	Greedy	42	0	84	23	0.658	14
Approve/block / 2	Search	44	0	88	32	0.686	19
Complexity time / 1	No review	0	0	0	0	0.000	52
Complexity time / 1	FCFS	50	12	56	22	0.442	18
Complexity time / 1	EDF	51	12	57	24	0.450	19
Complexity time / 1	Uncertainty	46	12	52	14	0.410	15
Complexity time / 1	Greedy	46	12	52	14	0.410	15
Complexity time / 1	Search	51	12	57	24	0.450	19
Complexity time / 2	No review	0	0	0	0	0.000	34
Complexity time / 2	FCFS	43	8	91	32	0.714	18
Complexity time / 2	EDF	43	8	91	32	0.714	19
Complexity time / 2	Uncertainty	41	11	87	26	0.686	14
Complexity time / 2	Greedy	41	11	87	26	0.686	14
Complexity time / 2	Search	43	7	90	31	0.706	19

at two, leaving service loss four, then approves the exchange. The unreviewed correct cancellation posts at its cutoff without error. This illustrates how preserving an opportunity under a public expected-value model can fail to preserve an actual error.

The first tie is bundle 00, replicate 0. Both policies approve the cancellation at two and block an incomplete return proposal at four. The modification has no staged proposal. Both have loss eight, consisting of the unresolved return and the unstaged modification. The first preparation failure is that modification, train:421. It finishes without a mutation; the failure field is null because no runtime exception occurs. This is still an unsuccessful retained workflow.

6.2 Saved-Data Benefit and Failure

The post hoc cohort contains a first qualifying benefit at bundle 28, replicate 0, with search-minus-greedy loss -4 . Its first unfavorable case is bundle 09, replicate 0, with difference $+4$, and its first tie is bundle 00, replicate 0. The first unstaged failure is train:062 in bundle 00, replicate 2, which exhausts all 14 tool turns. The complete source intentions, proposed and annotated actions, and tick-by-tick review and posting events are saved in the companion trace files. These examples are selected by first qualification, not effect size.

Table 5: Fresh cases. Search-minus-comparator loss differences. Negative favors search. Intervals resample bundles; W/T/L are scenario-bundle wins, ties and losses. Only the fresh approve/block two-tick search-greedy row is primary.

Condition	Comparator	Base	Difference	95% interval	W/T/L
Correction	greedy	1	0.000	[0.000, 0.000]	0/8/0
Correction	edf	1	0.000	[0.000, 0.000]	0/8/0
Correction	greedy	2	2.333	[0.000, 6.333]	0/6/2
Correction	edf	2	0.333	[0.000, 1.000]	0/7/1
Approve/block	greedy	1	0.000	[0.000, 0.000]	0/8/0
Approve/block	edf	1	0.000	[0.000, 0.000]	0/8/0
Approve/block	greedy	2	1.667	[0.000, 4.667]	0/6/2
Approve/block	edf	2	0.167	[0.000, 0.500]	0/7/1
Complexity time	greedy	1	0.000	[0.000, 0.000]	0/8/0
Complexity time	edf	1	0.000	[0.000, 0.000]	0/8/0
Complexity time	greedy	2	2.333	[0.000, 6.333]	0/6/2
Complexity time	edf	2	0.333	[0.000, 1.000]	0/7/1

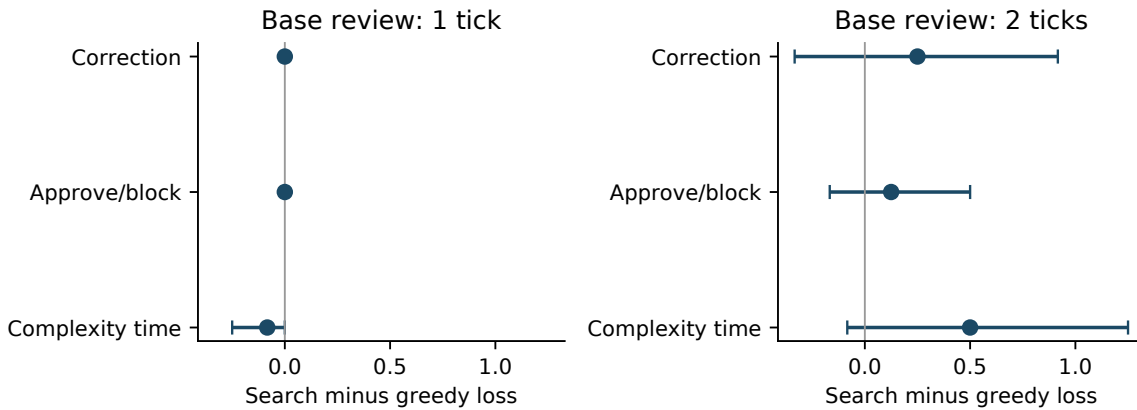


Figure 2: Post hoc saved-preparation sensitivity. The 32 bundles are separate from the eight fresh bundles and are not pooled for inference.

7 REPLAY AND REPRODUCTION

From a clean checkout, saved-data analysis requires Python 3.8 or later, NumPy and Matplotlib. It makes no inference requests. The declared inference path is one-shot, deadline-guarded and separate from analysis. Reproducing a fresh model output is not guaranteed by its seed. In earlier maintenance evidence, 38 identical-request groups had mixed actions; fresh generation variation and changes in feedback-dependent prompts are separate phenomena.⁷

The short saved-data path is:

```
bash scripts/replay_stage6.sh
```

This command verifies evidence, replays both cohorts, regenerates diagnostic tables and figures, and builds both PDFs. New analyses write only under the Stage 6 root. Historical reference traces are checked against their original values and all historical artifact hashes are preserved. The manuscript and supplement

⁷Drafting assistance: OpenAI Codex (OpenAI, 2026). See the AI assistance disclosure.

use the unchanged official template class, style and bibliography style. Source and build details are in the reproduction guide and stage report.

Previously disclosed Stage 5 redactions remove unsolicited credential-like model text from auxiliary arguments and later prompt histories. Original request, response and file hashes remain, with exact originals retained on the authorized pod and in an ignored local evidence directory. Transactions, states, labels and accounting do not change. Public redacted prompts are not claimed to be byte-identical regeneration inputs. Any new follow-up publication redaction is separately recorded under its own evidence root.

AI ASSISTANCE

OpenAI Codex assisted code, analysis, source checking, figure code and manuscript drafting and revision throughout this companion (OpenAI, 2026). Reported numeric evidence comes from retained model

Table 6: Post hoc Stage 5 preparations. Absolute outcomes across three generation replicates per bundle. Totals are repeated-condition counts, not independent samples. S and W denote service and wrong-posting loss. B is blocked unresolved requests and U is unstaged failures.

Condition / base	Policy	Loss	Mean	S	W	Correct	Wrong	B	U	Missed
Correction / 1	No review	800	8.333	424	376	182	51	0	55	51
Correction / 1	FCFS	220	2.292	220	0	233	0	0	55	0
Correction / 1	EDF	220	2.292	220	0	233	0	0	55	0
Correction / 1	Uncertainty	220	2.292	220	0	233	0	0	55	0
Correction / 1	Greedy	220	2.292	220	0	233	0	0	55	0
Correction / 1	Search	220	2.292	220	0	233	0	0	55	0
Correction / 2	No review	800	8.333	424	376	182	51	0	55	51
Correction / 2	FCFS	480	5.000	312	168	210	23	0	55	23
Correction / 2	EDF	440	4.583	292	148	215	18	0	55	18
Correction / 2	Uncertainty	332	3.458	260	72	223	10	0	55	10
Correction / 2	Greedy	332	3.458	260	72	223	10	0	55	10
Correction / 2	Search	356	3.708	264	92	222	11	0	55	11
Approve/block / 1	No review	800	8.333	424	376	182	51	0	55	51
Approve/block / 1	FCFS	424	4.417	424	0	182	0	51	55	0
Approve/block / 1	EDF	424	4.417	424	0	182	0	51	55	0
Approve/block / 1	Uncertainty	424	4.417	424	0	182	0	51	55	0
Approve/block / 1	Greedy	424	4.417	424	0	182	0	51	55	0
Approve/block / 1	Search	424	4.417	424	0	182	0	51	55	0
Approve/block / 2	No review	800	8.333	424	376	182	51	0	55	51
Approve/block / 2	FCFS	592	6.167	424	168	182	23	28	55	23
Approve/block / 2	EDF	572	5.958	424	148	182	18	33	55	18
Approve/block / 2	Uncertainty	496	5.167	424	72	182	10	41	55	10
Approve/block / 2	Greedy	496	5.167	424	72	182	10	41	55	10
Approve/block / 2	Search	508	5.292	424	84	182	11	40	55	11
Complexity time / 1	No review	800	8.333	424	376	182	51	0	55	51
Complexity time / 1	FCFS	284	2.958	240	44	228	5	0	55	5
Complexity time / 1	EDF	220	2.292	220	0	233	0	0	55	0
Complexity time / 1	Uncertainty	244	2.542	228	16	231	2	0	55	2
Complexity time / 1	Greedy	244	2.542	228	16	231	2	0	55	2
Complexity time / 1	Search	236	2.458	224	12	232	1	0	55	1
Complexity time / 2	No review	800	8.333	424	376	182	51	0	55	51
Complexity time / 2	FCFS	512	5.333	324	188	207	26	0	55	26
Complexity time / 2	EDF	464	4.833	300	164	213	20	0	55	20
Complexity time / 2	Uncertainty	332	3.458	260	72	223	10	0	55	10
Complexity time / 2	Greedy	332	3.458	260	72	223	10	0	55	10
Complexity time / 2	Search	380	3.958	272	108	220	13	0	55	13

outputs and deterministic simulation. Human authors must review and take responsibility for the complete submission. No human expert validation, author identity, funding or conflict declaration is invented.

Table 7: Post hoc Stage 5 preparations. Review resources. Time and waiting are summed abstract ticks. Utilization is mean occupied fraction of the processing horizon. More reviews do not necessarily imply more useful interventions.

Condition / base	Policy	Reviews	Corrections	Time	Waiting	Utilization	Multi-eligible
Correction / 1	No review	0	0	0	0	0.000	301
Correction / 1	FCFS	233	51	233	115	0.459	115
Correction / 1	EDF	233	51	233	115	0.459	115
Correction / 1	Uncertainty	224	51	224	97	0.442	115
Correction / 1	Greedy	224	51	224	97	0.442	115
Correction / 1	Search	233	51	233	115	0.459	115
Correction / 2	No review	0	0	0	0	0.000	212
Correction / 2	FCFS	182	28	364	152	0.712	87
Correction / 2	EDF	190	33	380	191	0.740	103
Correction / 2	Uncertainty	169	41	338	104	0.664	83
Correction / 2	Greedy	169	41	338	104	0.664	83
Correction / 2	Search	190	40	380	184	0.740	103
Approve/block / 1	No review	0	0	0	0	0.000	301
Approve/block / 1	FCFS	233	0	233	115	0.459	115
Approve/block / 1	EDF	233	0	233	115	0.459	115
Approve/block / 1	Uncertainty	224	0	224	97	0.442	115
Approve/block / 1	Greedy	224	0	224	97	0.442	115
Approve/block / 1	Search	233	0	233	115	0.459	115
Approve/block / 2	No review	0	0	0	0	0.000	212
Approve/block / 2	FCFS	182	0	364	152	0.712	87
Approve/block / 2	EDF	190	0	380	191	0.740	103
Approve/block / 2	Uncertainty	169	0	338	104	0.664	83
Approve/block / 2	Greedy	169	0	338	104	0.664	83
Approve/block / 2	Search	190	0	380	182	0.740	103
Complexity time / 1	No review	0	0	0	0	0.000	290
Complexity time / 1	FCFS	219	46	239	107	0.472	106
Complexity time / 1	EDF	233	51	256	133	0.506	115
Complexity time / 1	Uncertainty	216	49	237	104	0.467	109
Complexity time / 1	Greedy	216	49	237	104	0.467	109
Complexity time / 1	Search	232	50	254	132	0.502	114
Complexity time / 2	No review	0	0	0	0	0.000	200
Complexity time / 2	FCFS	175	25	365	144	0.714	86
Complexity time / 2	EDF	186	31	386	177	0.755	98
Complexity time / 2	Uncertainty	161	41	337	88	0.664	79
Complexity time / 2	Greedy	161	41	337	88	0.664	79
Complexity time / 2	Search	186	38	386	170	0.755	98

Table 8: Post hoc Stage 5 preparations. Search-minus-comparator loss differences. Negative favors search. Intervals resample bundles; W/T/L are scenario-bundle wins, ties and losses. Only the fresh approve/block two-tick search-greedy row is primary.

Condition	Comparator	Base	Difference	95% interval	W/T/L
Correction	greedy	1	0.000	[0.000, 0.000]	0/32/0
Correction	edf	1	0.000	[0.000, 0.000]	0/32/0
Correction	greedy	2	0.250	[-0.333, 0.917]	1/29/2
Correction	edf	2	-0.875	[-2.000, 0.000]	3/29/0
Approve/block	greedy	1	0.000	[0.000, 0.000]	0/32/0
Approve/block	edf	1	0.000	[0.000, 0.000]	0/32/0
Approve/block	greedy	2	0.125	[-0.167, 0.500]	1/29/2
Approve/block	edf	2	-0.667	[-1.417, -0.083]	4/28/0
Complexity time	greedy	1	-0.083	[-0.250, 0.000]	1/31/0
Complexity time	edf	1	0.167	[0.000, 0.500]	0/31/1
Complexity time	greedy	2	0.500	[-0.083, 1.250]	1/28/3
Complexity time	edf	2	-0.875	[-2.000, 0.000]	3/29/0

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